When and How to Use Derivative Action in a PID Controller

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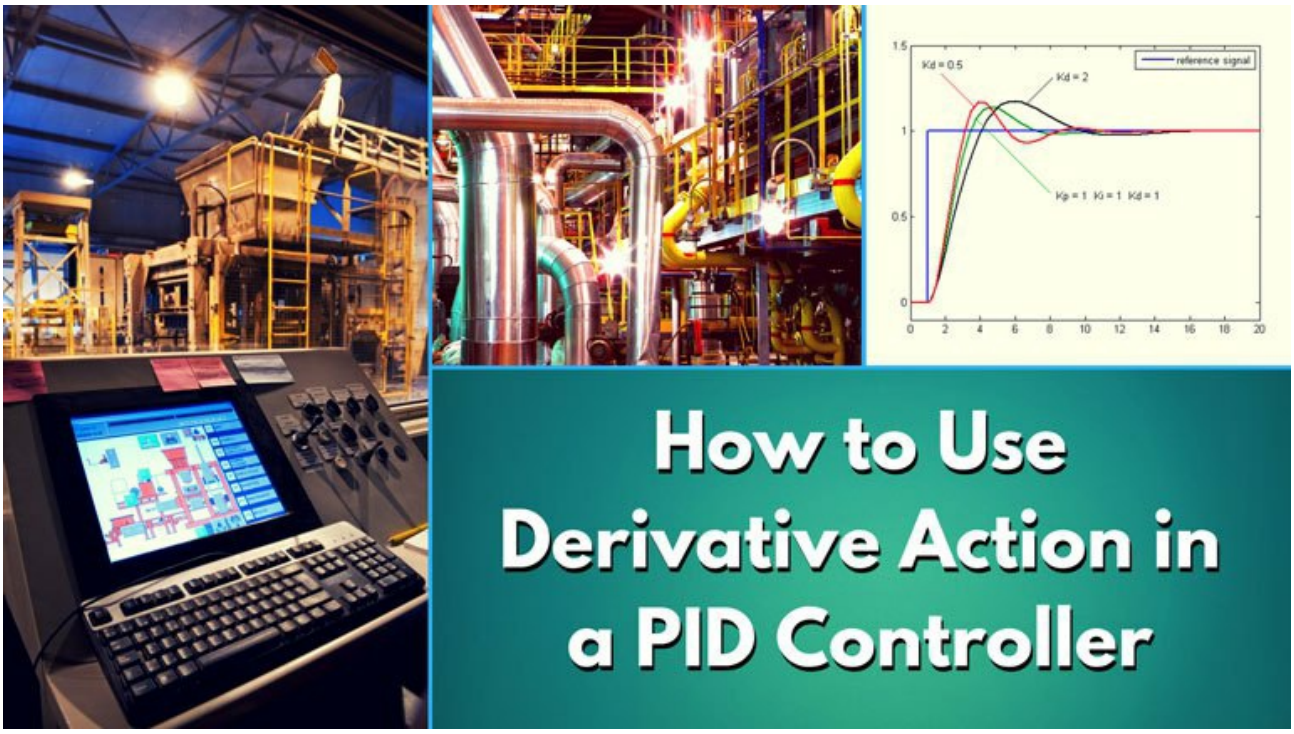
The following technical discussion is part of an occasional series showcasing the *ISA Mentor Program, authored by Greg McMillan, industry consultant, author of numerous process control books, 2010 ISA Life Achievement Award recipient and retired Senior Fellow from Solutia Inc (now Eastman Chemical). Greg will be posting questions and responses from the ISA Mentor Program, with contributions from program participants.*

Introduction

Derivative action is the least frequently used mode in the PID controller. Some plants do not like to use derivative action at all because they see abrupt changes in PID output and lack an understanding of benefits and guidance on how to set the tuning parameter (rate time). Here we have a question from one of the original protégés of the ISA Mentor Program and answers by a key resource on control Michel Ruel concluding with my view.

Hector Torres' Initial Question

Is there a guideline in terms of when to enable the derivate term in a PID?



Michel Ruel's Initial Answer

Derivative is more useful when dead time is not pure dead time but instead a series of small time constants; using derivative “eliminate” one of those small time constants.

You should use the derivative time equal to the largest of those small time constants. Since we usually do not know the details, a good rule of thumb is adjusting Derivative time to half the dead time.

Adding derivative (D) will increase robustness (higher gain and phase margin) since D will reduce apparent dead time of the closed loop.

A good example is the thermowell in a temperature loop: if the thermowell represents a time constant of 10 s, using a D of 10 seconds will eliminate the lag of the thermowell.

Hence, the apparent dead time of the closed loop is reduced and you can use more propositional, shorter integral time; the settling time will be shorter and stability better.

When you look at formulas to reject a disturbance, you observe that in presence of D, proportional and integral can be stronger.

We recommend using derivative only if the derivative function contains a built-in filter to remove high frequency noise. Most DCSs and PLCs have this function but some do not or there is a switch to activate the derivative filter.

Hector Torres’ Subsequent Question

What does having a higher phase margin increase the robustness?

Michel Ruel’s Subsequent Answer

Robustness means that the control loop will remain stable even if the model changes. Phase and gain margin represents the amplitude of the change before it becomes unstable, i.e. before reaching -180 degrees or a loop gain above one.

Ta analyze, we use open loop frequency response, the product of controller model and process model. On a Bode plot, gain are multiplied (or added if plot in dB) and total phase is the sum of process phase and controller phase.

Phase margin is the number of degrees required to reach -180 degrees when the open loop gain is 1 (0 dB). If this number is large (high phase margin), the system is robust meaning that the apparent dead time can increase without reaching instability. If the phase margin is small, a slight change in apparent dead time will bring the control loop to instability.

Adding derivative adds a positive phase, hence increases phase margin (compare to adding a dead time or a time constant that reduces the phase margin).

The ISA Mentor Program enables young professionals to access the wisdom and expertise of seasoned ISA members, and offers veteran ISA professionals the chance to share their wisdom and make a difference in someone’s career. Click this link to learn more about how

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Greg's Concluding Remarks

The use of derivative is more important in lag dominant (near-integrating), true integrating, and runaway processes (highly exothermic reactions). The derivative action benefit declines as the primary time constant (largest lag) approaches the dead time because the process changes become too abrupt due to lack of a significant filtering action by a process time constant.

Temperature loops have a large secondary time constant courtesy of heat transfer lags in the thermowell or the process heat transfer areas. Setting the derivative time equal to the largest of the secondary lags can cancel out almost 90 percent of the lag assuming the derivative filter is about 1/8 to 1/10 the rate time setting. Highly exothermic reactors can have positive feedback that causes acceleration of the temperature. Some of these temperature loops have only proportional and derivative action because integral action is viewed as unsafe.

If a PID Series Form is used, increasing the rate time reduces the integral mode action (increases the effective reset time), reduces the proportional mode action (decreases effective PID gain or increases effective PID proportional band) and moderates the increase in derivative action. The interaction factors moderates all of the modes preventing the resulting effective rate time from being greater than one-quarter the effective reset time. This helps prevent instability if the rate time setting approaches the reset time setting. There is no such inherent protection in the ISA Standard Form. It is critical that the user prevent the rate time from being larger than one-quarter the reset time in the ISA Standard Form. While in general it is best to identify multiple time constants, a general rule of thumb I use is the rate time should be the largest of a secondary time constant identified or one-half the dead time and never larger than one-quarter the reset time.

It is critical to convert tuning based on setting units and PID form used as you go from one vintage or supplier to another. It is best to verify the conversion with the supplier of the new system. The general rules for converting from different PID forms are given in the ISA Mentor Program Q&A blog post How Do You Convert Tuning Settings of an Independent PID with the last series of equations K1 thru K3 showing how to convert from a series PID form to the ISA Standard Form.

In general, PID structures should have derivative action on the process variable and not error unless the resulting kick in the PID output upon a setpoint change is useful to get to setpoint faster particularly if there is a significant control valve or VFD deadband or resolution limit.

A small setpoint filter in the analog output or secondary loop setpoint along with external reset feedback of the manipulated variable can make the kick a bump. A setpoint lead-lag on the primary loop where the lag time is the reset time and the lead is one-quarter of the lag or a two degrees of freedom structure with the beta set equal to 0.5 and the gamma set equal to about 0.25 can provide a compromise where the kick is moderated while getting to the primary setpoint faster.

See the ISA book *101 Tips for a Successful Automation Career* that grew out of this Mentor Program to gain concise and practical advice. See the *InTech* magazine feature article "Enabling new automation engineers" for candid comments from some of the original program participants. See the Control Talk column "How to effectively get engineering knowledge" with the ISA Mentor Program protégée Keneisha Williams on the challenges faced by young engineers today, and the column "How to succeed at career and project migration" with protégé Bill Thomas on how to make the most out of yourself and your project. Providing discussion and answers besides Greg McMillan and co-founder of the program Hunter Vegas (project engineering manager at Wunderlich-Malec) are resources Brian Hrankowsky (consultant engineer at a major pharmaceutical company), Michel Ruel (executive director, engineering practice at BBA Inc.), Leah Ruder (process systems automation group manager at the Midwest Engineering Center of Emerson Automation Solutions), Nick Sands (ISA Fellow and Manufacturing Technology Fellow at DuPont) and Bart Propst (Process Control Leader for the Ascend Performance Materials Chocolate Bayou plant).
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